

Trainings ▾

General Information

The intake manifold temperature and intake manifold pressure sensors are combined into a single sensor. The sensor is located in the intake manifold. Use the following procedure for a detailed component location view. Refer to Procedure 100-002 in Section E. (</qs3/pubsys2/xml/en/procedures/35/35-100-002-tr.html>)



Initial Check

Use an electronic service tool to monitor the value of the intake manifold pressure/temperature sensor with the key ON and the engine **not** running.

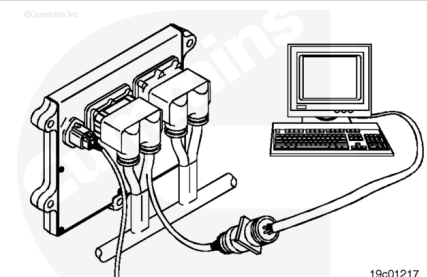
The value of the intake manifold temperature sensor should be checked when the engine is cold, and should read within 5.5°C or 10°F of the local ambient air temperature.

The value of the intake manifold pressure should read 0 kPa +/- 24 kPa [0 in Hg +/- 7 in Hg].

If either value is out of specification, first verify that the correct intake manifold pressure/temperature sensor is installed on the engine.

Compare the part number of the intake manifold pressure/temperature sensor to the part number listed in the parts catalog in QuickServe™ Online for the engine serial number.

If the correct part is installed, and either value is still out of specification, replace the intake manifold pressure/temperature sensor.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

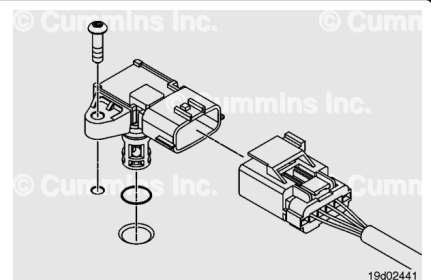


- Disconnect the batteries. Refer to the original equipment manufacturer (OEM) service manual.
- Clean the area around the intake manifold pressure/temperature sensor.

Remove

⚠ CAUTION ⚠

Be careful not to damage the o-ring seal when removing the sensor. Do not pry the sensor, as damage to the plastic cage can occur.



Disconnect the sensor connector from the engine harness.

Remove the mounting capscrew.

Remove the sensor from the engine by pulling straight up on the sensor.

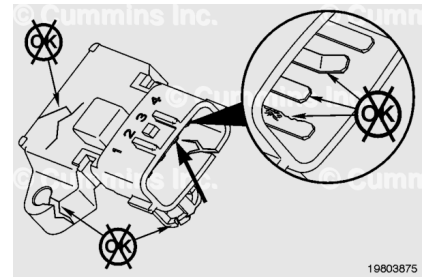
Clean and Inspect for Reuse

Inspect the engine harness connector and the intake manifold pressure/temperature sensor for the following:

- Cracked or broken shell
- Missing or damaged connector seals
- Dirt, debris, or moisture in or on the connector pins

- Corroded, bent, broken, pushed back, or expanded pins
- Chipped, cracked, extruded, or damaged sensor.

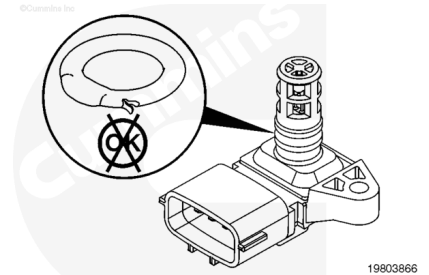
Repair or replace parts as necessary.



Inspect the intake manifold pressure/temperature sensor for the following:

- Swollen o-ring
- Nicks or cuts in or on the o-ring.

Replace the o-ring if **any** damage is found.



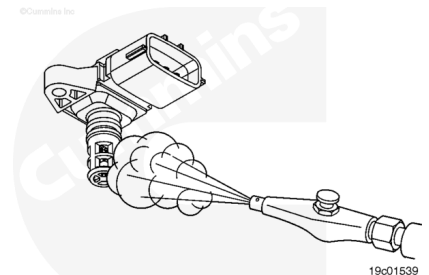
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Inspect the tip of the sensor for soot and carbon buildup.

Use compressed air to remove the soot from the sensor, if necessary.

Do **not** replace the sensor for soot in the sensor.



Install

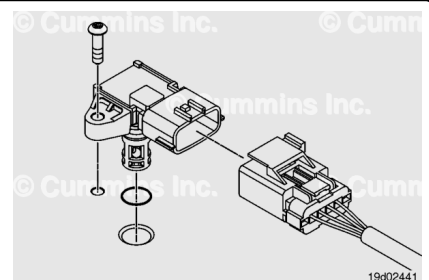
Lubricate the o-ring with clean engine oil before installation.

Install the intake manifold pressure/temperature sensor by pressing firmly on the top of the sensor until the o-ring is fully seated.

Install and tighten the capscrew.

Torque Value: 7 n•m [62 in-lb]

Connect the engine harness to the intake manifold pressure/temperature sensor.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries. Refer to the OEM service manual.
- Operate the engine and check for leaks.

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